

Case Report

Pyrolysis of PP and HDPE from plastic packaging waste into liquid hydrocarbons using natural zeolite Lampung as a catalyst

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ABSTRACT

This study aims to convert plastic waste into fuel oil through the pyrolysis process that assisted by natural zeolite as catalyst. Evaluation of the catalytic ability of natural zeolite was carried out in a batch reactor with various reaction conditions. The optimum conditions for pyrolysis of PP and HDPE samples were obtained by adding a catalyst of 7% and 3% (w/w) with an oil yield of 69.69% and 65.60% liquid fraction, respectively. The use of natural zeolite catalysts in the pyrolysis of PP and HDPE from plastic waste offered a simple method and demonstrated reliable catalyst performance.

1. Introduction

The increase in the use of plastic is a consequence of the development of technology, industry, and population [1]. Plastic has many benefits in daily human needs such as packaging, construction, electronics, and even medicine [2,3]. Compared to other materials, plastic's advantages include being strong, lightweight, flexible, rust-resistant, not easily broken, easy to color, easy to shape, and good heat and electrical insulators [4–7]. However, the high use of plastic causes new problems for the environment. In 2020, Indonesia produced up to 32.82 tons of plastic waste, which is estimated to continue to increase. This plastic waste can be dangerous in various aspects, including increasing the death of various soil invertebrates, inhibiting growth and reducing the rate of plant photosynthesis that enters through the roots (micro-/nano-plastic) to be eaten by humans and animals, and affects aquatic organisms [8]. The problem is that plastic waste is difficult to decompose, absorb water, and cannot rust, so it is very difficult to be degraded by nature [9–11].

Today's plastic waste is often just landfilled, burned, or recycled. This process has not solved all plastic waste problems because the landfill has been unable to decompose plastic waste [12,13]. If plastic waste is burned at low temperatures, it will produce harmful carcinogenic compounds such as poly chloro-dibenzodioxins and poly chloro-dibenzofurans [14–16]. So currently, ongoing research is being carried out to deal with this problem. One approach that is interesting

and easy to do for the chemical recycling of plastic waste is pyrolysis. Pyrolysis of plastic waste will break down large polymer molecules into smaller chains so they can be used as feedstocks such as biofuels after going through an advanced process to get the composition to the legal limitations [17–19]. The breakdown of these large chain polymers occurs at high temperatures in an inert gas environment [20–22]. The product formed from plastic pyrolysis is divided into solid, liquid, and gas phases. Liquid products that can be used as fuel usually consist of a mixture of naphthenes, olefins, paraffins, and some aromatic compounds [23–25]. The formation of fuels from plastics is more profitable by pyrolysis than other processes such as combustion, gasification, and liquefaction [26,27]. Pyrolysis processes that only utilize heat energy often result in low product yields. This approach requires high temperatures from 300°C to 900°C [28,29]. So, it is often conducted with a catalyst assistance approach in the reaction.

Using a catalyst to help run the reaction is the right choice because it offers a faster process at a temperature that is not high [30,31]. Previous research has used several methods, such as catalytic cracking, hydro-treating, and esterification [32,33]. Various types of catalysts have been tested, ranging from homogeneous to heterogeneous catalysts, such as FCC, metal oxides, acidic and non-acidic mesoporous materials, to different other composites [34–38]. For example, Aboul-Enein et al. used a Fe-Mo/MgO composite catalyst to pyrolyze polyethylene samples and produces high yields [39].

The use of zeolite as a catalyst is not something new. Several previous

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studies show that zeolite has high selectivity in catalytic cracking of paraffin raw materials, which is influenced by the pores it has [40,41]. In addition, the use of zeolite as a catalyst can reduce the use of less energy (low reaction temperature) compared to the use of metal oxide catalysts [42]. Zeolites have high catalytic activity, are not easy to agglomerate, have large porosity, are stable at high temperatures, and are abundant in Indonesia [43,44]. In the plastic pyrolysis reaction, the pore structure and acidity of the zeolite are suitable for increasing liquid and gaseous products. Chen et al. used a ZSM-5 zeolite catalyst in LDPE pyrolysis which was highly selective to produce high-valued light aromatic and olefins [45]. Dai et al. used a modified ZSM-5 catalyst for cracking of HDPE and demonstrated that the density of the Brønsted acid sites impacted decreasing catalyst lifetime [46].

In this study, natural zeolite obtained from Lampung (Indonesia) was used as a catalyst in the pyrolysis process of polypropylene (PP) and high-density polyethylene (HDPE) from plastic packaging waste into liquid hydrocarbons. As a sample, plastic waste packaging instant noodles that are widely available in Indonesia were chosen as a source of PP and clear plastic bag waste as a source of HDPE. Processing of plastic waste through pyrolysis with natural zeolite catalysts offers simple preparation and low prices and will reduce the number of plastic pollutants.

2. Experimental procedures

2.1. Materials

All the chemicals and reagents were used for analytical grade without further purification. Some of the materials used are polypropylene (PP) instant noodle packaging waste, high-density polyethylene (HDPE) plastic packaging waste, NH_4Cl 1 M (Merck), AgNO_3 (Merck), acetone (Merck), distilled water, and natural zeolite from Lampung Province, Indonesia.

2.2. Activation of natural zeolite Lampung

Clinoptilolite natural zeolite was ground and sieved with a size of 60 mesh. Furthermore, the samples were immersed in distilled water while stirring for three days (8 hours/day) at room temperature. The precipitate formed was filtered and dried in an oven at a temperature of 120°C for 2 h. Furthermore, the zeolite was immersed in NH_4Cl 1 M solution for 24 hours at room temperature. After completion, the zeolite was filtered, washed with distilled water and tested for chloride ion content using silver nitrate (AgNO_3) solution until there was no precipitate. This is done to purify the zeolite from Cl^- ions due to the addition of NH_4Cl . The zeolite was dried in an oven at a temperature of 120°C . After cooling, zeolite was placed in a porcelain cup and calcined for 8 hours at 450°C in a furnace.

2.3. Pyrolysis of PP and HDPE from plastic packaging waste

PP and HDPE plastic samples in different containers were cleaned and cut into small pieces. The use of PP plastic samples from waste packaging instant noodles and clear plastic bag waste as a source of HDPE is because these types of plastics are very much found in Indonesia as waste due to the large amount of use. The pyrolysis process was carried out using a batch reactor made of solid cylindrical iron with an inner diameter of 5 cm, an outer diameter of 7 cm, and a height of 15 cm. Plastic samples (20 g) were put into the reactor and added with variations in the concentration of zeolite catalyst (1%, 3%, 5%, 7%, and 9% w/w). The reactor heater was run from room temperature to 400°C so that the cracking or pyrolysis process occurs and the pyrolysis product is obtained in the reservoir flask. The reaction was carried out until the last drop of the liquid fraction product was obtained. The series of tools in the pyrolysis process of PP and HDPE plastic samples are shown in Fig. 1. The yield of the reaction is determined by calculation using the formula:

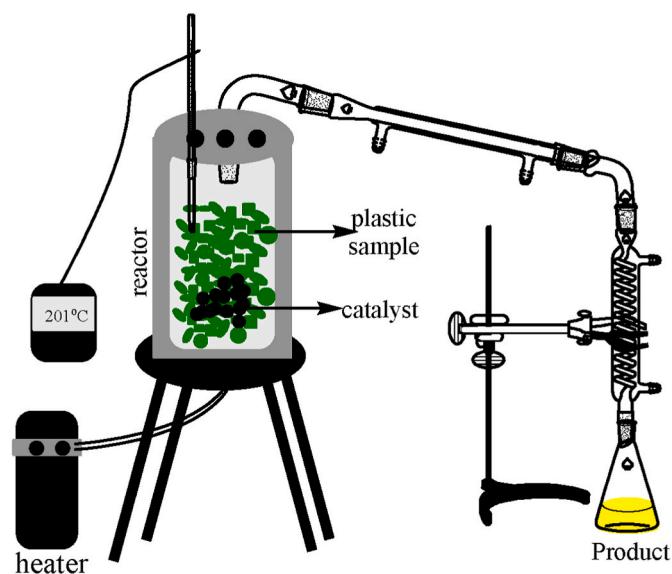


Fig. 1. The pyrolysis reactor equipment schematic.

$$\text{Yield} = \frac{\text{Mass of liquid phase}}{\text{Mass of sample}} \times 100\%$$

3. Results and discussions

In nature, zeolites are usually bound with alkaline or alkaline earth such as Ca^{2+} , Na^+ , and K^+ (some cases are Mg^{2+} , Ba^{2+} , Li^+ , and Sr^{2+}) which are weakly bound by electrostatic bonds and are present on the outer surface of the zeolite [47,48]. The addition of NH_4^+ ions will replace cations (exchangeable cations) so that alkaline or alkaline earth ions in natural zeolite will be released and replaced by NH_4^+ ions [49]. An illustration of the ion change is shown in Fig. 2. The heating process will convert NH_4^+ into NH_3 vapor and leave H^+ bound to the zeolite to form H-zeolite. The loss of cations on the surface of the zeolite will provide more templates or pores, thereby increasing the catalytic activity. This process will also cause an increase in the number of Brønsted acid sites in the zeolite.

Natural zeolite before and after activation was characterized by XRD. The X-ray diffraction patterns of the sample are shown in Fig. 3. The major peaks found in the X-ray diffraction signal at 2θ are 9.8° ; 21.8° ; 22.4° ; 26.1° ; 28.1° ; 30.1° with d values 8.983; 4.058; 3.962; 3.417; 3.178; and 2.970, respectively of zeolite before activated. After activation, there was a shift and increase in the intensity of several peaks, especially at $2\theta = 27.7^\circ$; 28.1° ; 28.4° (Fig. 3a). These three peaks are characteristic of clinoptilolite crystals (ICDD No. 39-1383) which indicate the presence of potassium sodium calcium aluminum silicate hydrate minerals from clinoptilolite-Na, with the chemical formula $\text{KNa}_2\text{Ca}_2(\text{Si}_{29}\text{Al}_7)\text{O}_{72} \cdot 24\text{H}_2\text{O}$. The increase in peak intensity indicates an increase in the crystal purity of natural zeolite after activation which supports the release of impurities in the zeolite.

The cracking reaction or pyrolysis is a reaction to break C-C bonds from long carbon chains and large molecular weights to hydrocarbons with short carbon chains and small molecular weights (Fig. 4). PP and HDPE plastic samples were subjected to high temperatures to produce free radical fragments. At the initiation stage, polymers that are large molecules (1) will be broken down into free radicals (2), and continue to undergo splitting into smaller parts (3) during the propagation process. The formation of olefin hydrocarbons (4) can also occur during the propagation process. The cracking reaction will stop at the termination stage, where the radicals meet each other (5) and undergo polymerization again to form oligomers (6) [6,50,51]. The presence of a zeolite catalyst in this study played a role in the protonation of the polymer in a

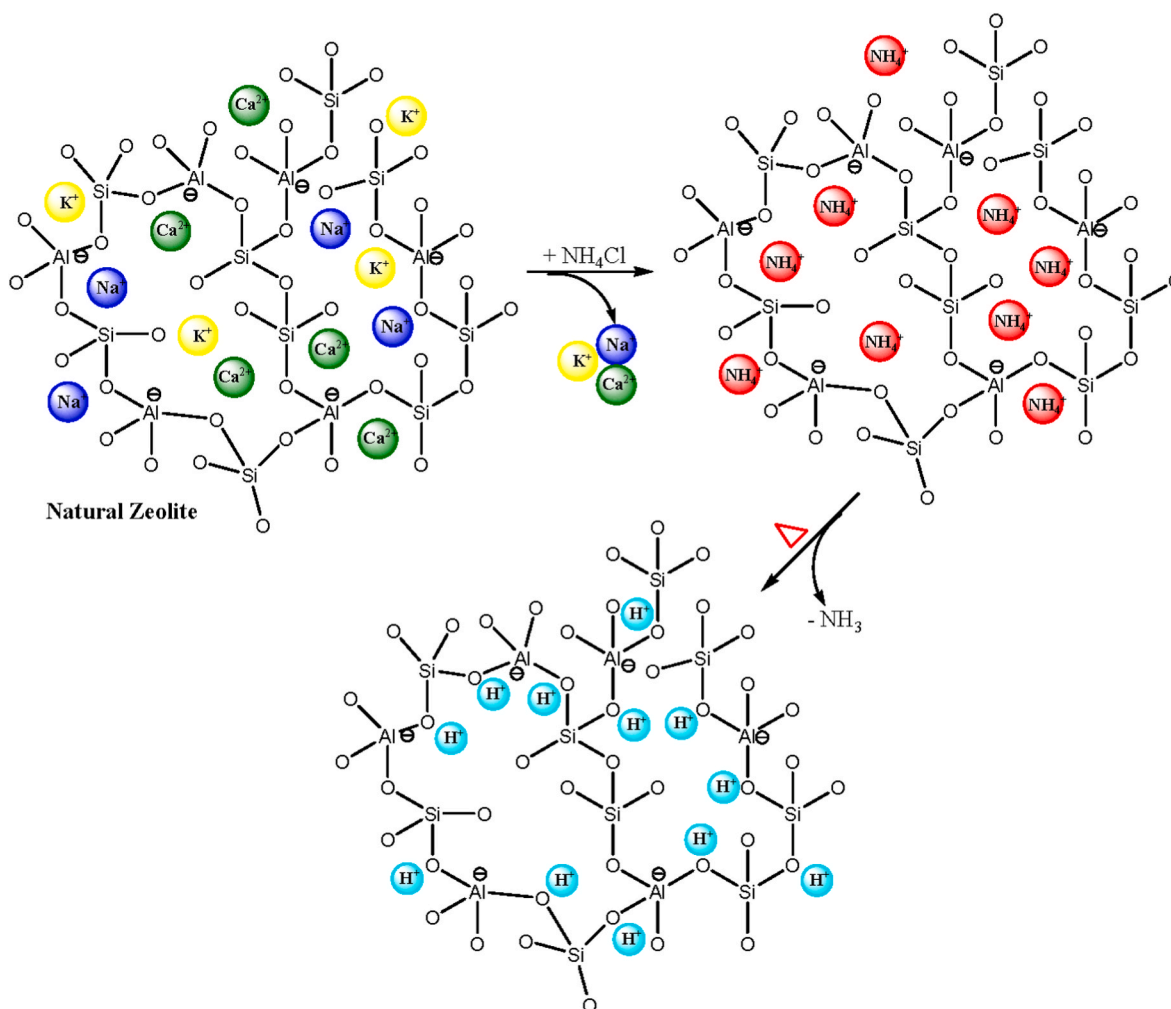


Fig. 2. The proposed activation and removal of cations in natural zeolite.

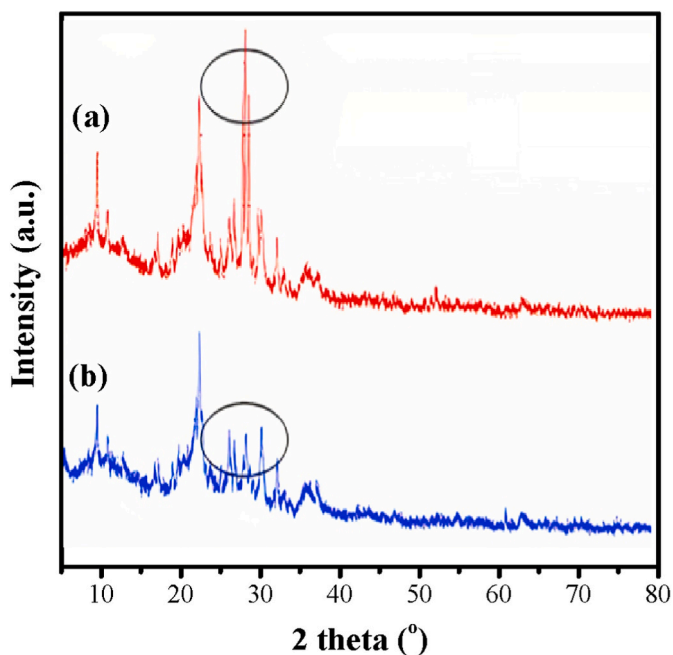


Fig. 3. Comparison of zeolite diffractogram after (a) and before activation (b).

saturated (single chain) bond with the acid side of the catalyst to form carbonium ions. The acidity value of the zeolite in this study was 0.4704 mmol NH_3/g which was obtained by the NH_3 -TPD method. This value is included in the medium group so that the ability of the catalyst should still be improved by increasing the acidity of the zeolite [52]. In addition, the Brønsted acid site on zeolite also results in reduction of oxygenated hydrocarbons and increases isomerization and saturation reactions [53,54]. This activity will increase with the support of high-temperature conditions so that it will increase the conversion results per unit of time compared to without a catalyst.

Pyrolysis products from PP and HDPE packaging waste are liquid, solid, and gas products. The liquid product that is the subject of this research is a yellow hydrocarbon liquid with a pungent odor. The yield obtained from the variation of reaction conditions is shown in Fig. 5. The addition of a catalyst in the pyrolysis process increases the desired liquid product by increasing the conversion of plastics. As shown in Fig. 5, the pyrolysis reaction without using a catalyst only produces a liquid hydrocarbon product of 29.40% for the PP sample and 12.02% for the HDPE sample. The addition of a natural zeolite catalyst resulted in a much higher product yield compared to the uncatalyzed reaction. At the use of 1% catalyst, the resulting liquid hydrocarbon is 67.09% for the PP samples and 46.65% for the HDPE samples. The highest yield of liquid product was obtained in the addition of 7% catalyst for PP samples and 3% catalyst for HDPE samples with values of 69.69% and 65.60%, respectively.

The use of temperature and reaction time during the sample pyrolysis process is shown in Table 1. The temperature range in Table 1 is the

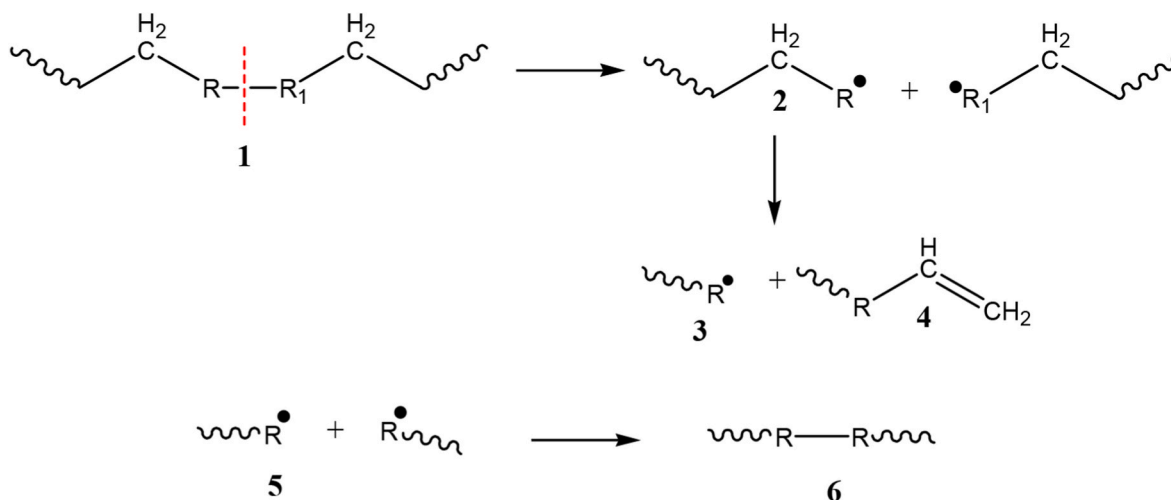


Fig. 4. The proposed mechanism of thermal pyrolysis of plastic samples.

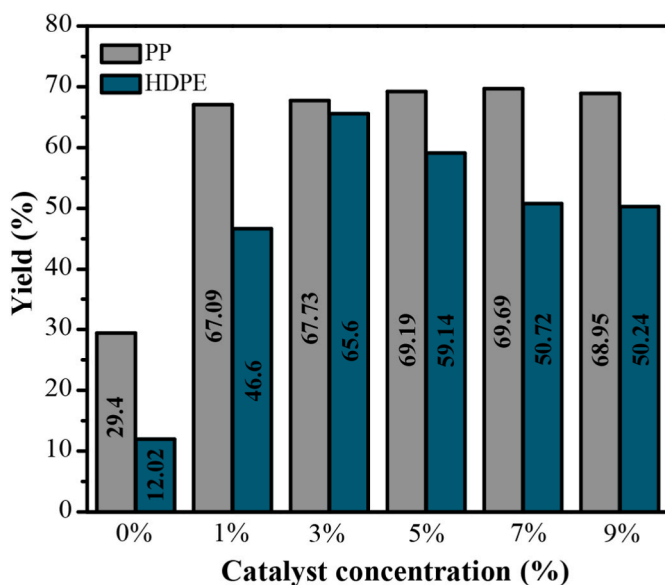


Fig. 5. The proposed mechanism of thermal pyrolysis of plastic samples.

Table 1

Pyrolysis reaction conditions of PP and HDPE plastic samples.

Catalyst concentration	Temperature range (°C)		Time reaction (minute)	
	PP	HDPE	PP	HDPE
0	284–330	243–340	149	192
1	270–329	230–315	140	189
3	251–330	224–309	130	132
5	195–313	223–304	110	138
7	179–311	197–240	105	138
9	212–325	207–270	115	154

temperature at the start of the liquid product drop to the last drop. Using a zeolite catalyst shows a decrease in the total time from the start of heating to the last drop of liquid hydrocarbon product. The first drop of liquid hydrocarbon product was obtained at a temperature of 284°C for the PP sample and 243°C for the HDPE sample. While under optimum conditions, the first drop of liquid product was obtained at 197°C for PP samples (7% catalyst) and 224°C for HDPE samples (3% catalyst). Similarly, the reaction time required for the reaction without a catalyst takes 149 minutes for the PP sample and 192 minutes for the HDPE

sample (rendered in Fig. 5). While using a catalyst, the time is shorter, namely 105 minutes for PP samples and 132 minutes for HDPE samples. With the high yield obtained and the use of lower temperatures and time, natural zeolite has good potential as a catalyst in the pyrolysis reaction of PP and HDPE plastic samples. Compared to other catalyst, this catalyst offers a simple process, easy-to-obtain natural zeolite utilization, and inexpensive way of pyrolysis of plastics that can be used as feedstocks such as biofuels. Besides that, this is also one of the efforts in processing plastic waste to reduce environmental harm.

The analysis of the functional group vibrations of the liquid hydrocarbon products resulting from the pyrolysis of PP and HDPE plastics was carried out by FTIR analysis. The FTIR spectrum of the PP and HDPE pyrolysis liquid products is shown in Fig. 6. The FTIR spectra of HDPE (Fig. 6a) and PP (Fig. 6b) products do not look much different. Several

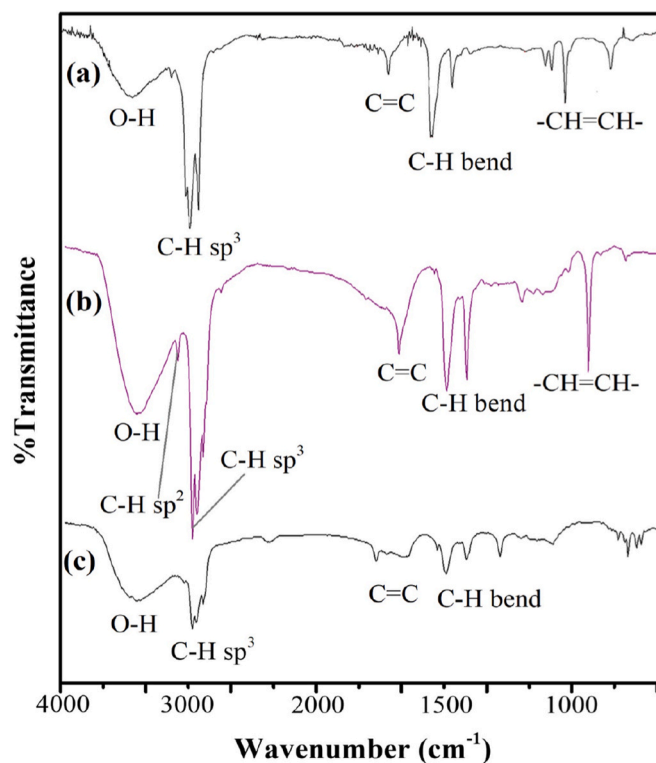


Fig. 6. FTIR spectra of the liquid hydrocarbon products resulting from the pyrolysis of (a) HDPE, (b) PP, and (c) commercial premium (RON 88).

functional groups were detected including the C-H sp^2 at $\sim 3026\text{ cm}^{-1}$ which was supported by the appearance of a peak in the range $1500\text{--}1600\text{ cm}^{-1}$ as the vibration of the C=C functional group. The presence of functional groups C-H sp^2 and C=C as a marker for forming olefins during the pyrolysis process. In addition, a stretching vibration of C-H sp^3 is at $\sim 2950\text{ cm}^{-1}$ and the bending vibration of C-H is spread at $1400\text{--}1500\text{ cm}^{-1}$ [55,56]. Compared to commercial fuel oil products (Premium, RON 88), it has the same functional group vibration absorption as the two liquid hydrocarbon products from PP and HDPE samples. In this study also analyzed the functional groups of solid products resulting from plastic pyrolysis of PP and HDPE samples, as shown in Fig. 7. Both products in HDPE (Fig. 7a) and PP (Fig. 7b) had similar functional group absorptions with liquid hydrocarbon products. .

The analysis of liquid hydrocarbon products from the pyrolysis of PP and HDPE plastic samples at optimum conditions was carried out by GCMS analysis. The distribution of liquid products is grouped based on the distribution of carbon chain lengths as shown in Table 2. The hydrocarbon derivative of the liquid product from the PP sample looks different from the HDPE sample. The distribution of liquid products resulting from PP plastic samples pyrolysis was dominated by C5-C12, followed by C17-C20, and the lowest was C13-C16 with values of 74.16%, 22.32%, and 3.52%, respectively. This indicates that the liquid product from the plastic pyrolysis of PP samples is primarily the light fraction or gasoline (C5-C12) and a small portion is the diesel fraction (C18-C28). In contrast to the liquid hydrocarbon product acyl pyrolysis of HDPE plastic samples, the distribution of the liquid product is on the carbon chain C13-C16 (40.39%), C5-C12 (30.36%), C17-C20 (24.69%), and >C20 (4.56%). About 70% of liquid hydrocarbon products are distributed in the gasoline and diesel range. The remaining liquid hydrocarbons C17-C21 formed 29%.

Measurement of the density of the liquid product from pyrolysis of PP and HDPE samples was carried out and compared with several groups of local fuel brands (Indonesian State-owned Oil and natural gas corporation) such as Pertamina (RON 92), Premium (RON 88), and Diesel fuel (CN 48) as shown in Table 3. The results show that the density of the liquid hydrocarbon product of PP is close to the density values of Pertamina and Premium fuels. In contrast, the liquid hydrocarbon product of HDPE is closer to those of diesel fuel.

4. Conclusion

In summary, the pyrolysis of waste plastic packaging polypropylene (PP) and high-density polyethylene (HDPE) has been successfully carried out with the help of natural zeolite catalysts from Lampung, Indonesia. Pyrolysis without catalyst only resulted in a small yield of liquid hydrocarbon product (PP: 29.40%, HDPE: 12.02%) while the use of zeolite catalyst gave a significant increase in yield (PP: 69.69%, HDPE: 65.60%). The use of natural zeolite catalysts also shortens the reaction time and temperature during the pyrolysis process. The distribution of liquid hydrocarbons from the pyrolysis products of PP samples was C5-C12 (74.16%), C17-C20 (22.32%), and C13-C16 (3.52%). Meanwhile, the distribution of liquid hydrocarbons from the pyrolysis products of HDPE samples was C13-C16 (40.39%), C5-C12 (30.36%), C17-C20 (24.69%), and >C20 (4.56%). It is worth mentioning that using zeolite catalysts offers good catalytic ability in producing liquid hydrocarbons from the pyrolysis of PP and HDPE plastic samples with high yields and shorter times. Also, the handling of plastic waste through the pyrolysis approach with other catalysts would be very interesting.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

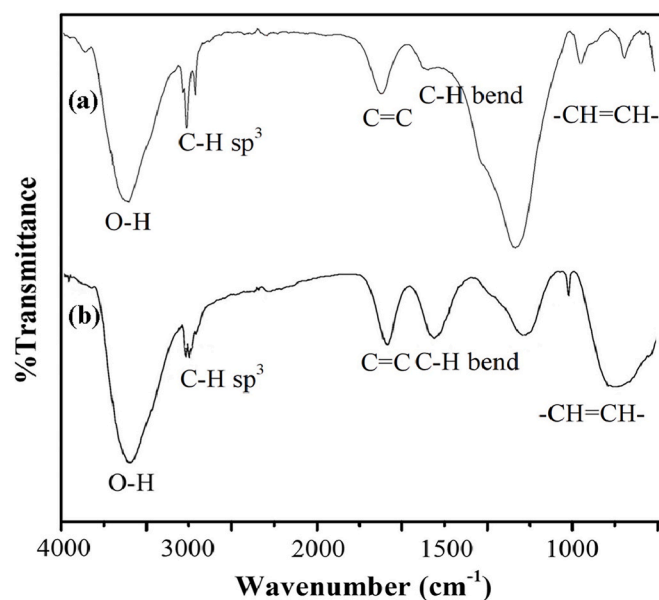


Fig. 7. FTIR spectra of the solid products resulting from the pyrolysis of (a) HDPE, and (b) PP.

Table 2

Distribution of pyrolysis liquid hydrocarbon products from PP and HDPE plastic samples.

Feed Material	Distribution of liquid hydrocarbon products (%)			
	C5-C12	C13-C16	C17-C20	>C20
Polypropylene	74.16	3.52	22.32	0
High-density polyethylene	30.36	40.39	24.69	4.56

Table 3

Density comparison of liquid hydrocarbon products with some commercial fuels.

Sample	Density (g/mL)
Liquid hydrocarbon products of PP	0.8085
Liquid hydrocarbon products of HDPE	0.8779
Premium (RON 88)	0.8031
Pertamax (RON 92)	0.8232
Diesel fuel (CN 48)	0.9154

Data availability

No data was used for the research described in the article.

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